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1) A) Because the data is already seen so it won't generalize.

B) use G to generate Dtest.
It would be beneficial if
$D_{test} \cap D_{train} = \emptyset$
If this is true, h(Dtest) has $0 \leq score \leq 1$. else $0 \leq score \leq 1$

C) It generalizes.

D) the score would be less.

2)

A) No; but in the second case they would be closer.

B) This is evaluating 1 h for 1 D_n, D_t
we would like to see performance across different h for diferrent D_n, D_t

C) No. This is still the same classifier.

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D) A better way would be Cross-Validation.
We would ~~T~~ 2 - partition G in T different ways. each into Dtest / Dtrain and get a new h for each.

```
def better_eval_learning_alg(L, G, n):
    for t in range(T):
        h ~~~~ , Dtrain = G(n)
        D_test^(t) = G(n)
        h^(t) = L(Dtrain)
    return 1/T Σ_{t=0}^{T} score(h^(t), D_test^(t))
```

$0 \leq \text{values} \leq 1$

E) This method evaluates the ability to learn h, Not the h.

A) Same way.